

Proposed Solution

Date	1 October 2026
Team ID	SWTID-2026-7957
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal describes EduGenie, a generative-AI learning assistant that helps students and self-learners study more efficiently. It tackles static content and information overload in today's learning tools, and its key features include instant question answering, simple explanations, practice quizzes, summaries and personalized learning paths.

Project Overview	
Objective	To help students and self-learners study faster and with more confidence by using generative AI to answer questions, explain concepts in simple language, create practice quizzes, summarize long text and build step-by-step learning paths in one simple web application.
Scope	A free, browser-based learning assistant with five features (Q&A, Explain, Summarize, Quiz and Learning Path), built on a FastAPI backend that combines the Google Gemini API with the small local LaMini-Flan-T5 model. The first release is open access with no login and stores no user data. Progress tracking, voice input, multilingual support and LMS integration are future scope.
Problem Statement	
Description	Learners face static study material, search results they must sift by hand, no easy way to test themselves, and no clear roadmap when they start a new subject. This wastes study time and leaves them overwhelmed and unsure of their level.
Impact	Solving these problems saves study time, reduces information overload, gives learners instant feedback on what they understand, and helps them stay motivated and prepared for exams.
Proposed Solution	
Approach	A FastAPI backend takes the learner's input from a simple web page and sends carefully written prompts to Google Gemini (and to the local LaMini-Flan-T5 model for concept explanations). It then cleans and checks the response, for example the quiz JSON, and returns a readable result to the page.

Key Features	• Question answering with short, accurate answers • Concept explanation in simple language • 3-question multiple-choice quiz with instant Correct / Incorrect feedback • Summaries of long passages for quick revision • Beginner-to-advanced learning path with timelines and resources • Responsive, no-login interface, with errors shown clearly in the page • API key kept in environment variables; live deployment on Vercel
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Any modern multi-core CPU (for example Intel i3/i5 or higher, or Apple M1). No GPU is required.
Memory	RAM specifications	4 GB minimum; 8 GB recommended for smoother local model loading.
Storage	Disk space for data, models and logs	About 10 GB free space for the Python libraries (including PyTorch), the local LaMini model files (about 3 GB) and the application.
Software		
Frameworks	Python frameworks	FastAPI (backend) with Jinja2 templates.
Libraries	Additional libraries	Uvicorn (ASGI server), Google GenAI SDK, Transformers, PyTorch, SentencePiece, python-dotenv.
Development Environment	IDE	Visual Studio Code (or PyCharm or any editor that supports Python 3.10+), with a Python virtual environment; Git and GitHub.
Cloud Services	APIs and hosting	Google AI Studio (Gemini API key) and Vercel (web hosting).
Data		
Data	Source, size, format	No dataset is used or stored. Inputs are short texts typed by the learner; outputs are plain text, Markdown and small JSON (quiz) generated by the AI models when requested. The LaMini model files (about 3 GB) are downloaded once from Hugging Face.